

Algebra 1 — Homework 2

Source: Tutorial sheet 2 (Tutorials/Tutorial 2.pdf) Opened 18 Aug 2026 Due Thu 20 Aug 2026, 11:59 PM

Submit: problems 2(a), 3, 4.

2(a). We have seen that $\text{RREF}(A)$ lets you write the solution space of $Ax = 0$ as the span of a “magic” set of vectors. Do this for

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 0 & 4 & 6 \\ 1 & 2 & 2 & 5 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

3. “The general solution of a consistent system is a translate of a subspace,” i.e. of the form $W + u$ with W a subspace (of what?) and u a vector. Demonstrate this explicitly for

$$\begin{pmatrix} 1 & -1 & 2 & 3 \\ 2 & -2 & 5 & 4 \\ -1 & 1 & -1 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \\ -4 \end{pmatrix}.$$

4. For an $r \times c$ matrix A , check that $f(x) = Ax$ from $\mathbb{R}^c$ to $\mathbb{R}^r$ is linear. Show that every linear map $L : \mathbb{R}^c \rightarrow \mathbb{R}^r$ arises this way: for a given L there is a unique matrix A with $L(x) = Ax$.